

OSSP SKILL

SKILL-1

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
rishitha@Rishitha: ~/system-  
rishitha@Rishitha:~$ sudo apt update  
[sudo: authenticate] Password:  
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease  
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease  
2 packages can be upgraded. Run 'apt list --upgradable' to see them.  
rishitha@Rishitha:~$ sudo apt upgrade -y  
Not upgrading yet due to phasing:  
python3-software-properties software-properties-common  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2  
rishitha@Rishitha:~$ sudo apt install build-essential -y  
build-essential is already the newest version (12.12ubuntu2.26.04.2).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2  
rishitha@Rishitha:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
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This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
rishitha@Rishitha:~$ make --version  
GNU Make 4.4.1  
Built for x86_64-pc-linux-gnu  
Copyright (C) 1988-2023 Free Software Foundation, Inc.  
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>  
This is free software; you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
rishitha@Rishitha:~$ mkdir system-programming  
mkdir: system-programming: File exists  
rishitha@Rishitha:~$ cd system-programming  
rishitha@Rishitha:~/system-programming$ pwd  
/home/rishitha/system-programming  
rishitha@Rishitha:~/system-programming$ mkdir src include bin  
mkdir: src: File exists  
mkdir: include: File exists  
mkdir: bin: File exists  
rishitha@Rishitha:~/system-programming$ touch Makefile  
rishitha@Rishitha:~/system-programming$ ls  
Makefile README.md bin include src  
rishitha@Rishitha:~/system-programming$ nano src/main.c  
rishitha@Rishitha:~/system-programming$ gcc src/main.c -o bin/main  
rishitha@Rishitha:~/system-programming$ ./bin/main  
Hello, Linux System Programming!  
rishitha@Rishitha:~/system-programming$ nano Makefile  
rishitha@Rishitha:~/system-programming$ make  
make: Nothing to be done for 'all'.  
rishitha@Rishitha:~/system-programming$ ./bin/main  
Hello, Linux System Programming!  
rishitha@Rishitha:~/system-programming$ make clean  
rm -f bin/main  
rishitha@Rishitha:~/system-programming$ nano src/fork_demo.c  
rishitha@Rishitha:~/system-programming$ gcc src/fork_demo.c -o bin/fork_demo  
rishitha@Rishitha:~/system-programming$ ./bin/fork_demo  
Parent process  
Child process  
Parent PID: 1259  
Child PID: 1260  
Child PID: 1260  
Parent PID: 1259  
rishitha@Rishitha:~/system-programming$ nano src/exec_demo.c  
rishitha@Rishitha:~/system-programming$ gcc src/exec_demo.c -o bin/exec_demo  
rishitha@Rishitha:~/system-programming$ ./bin/exec_demo  
Before exec()  
total 16
```

```

rishitha@Rishitha: ~/system-;
total 16
-rw-r--r-- 1 rishitha rishitha 155 Aug 17 16:50 Makefile
-rw-r--r-- 1 rishitha rishitha 0 Aug 17 16:45 README.md
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:51 bin
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:45 include
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:51 src
rishitha@Rishitha:~/system-programming$ ls -l
total 16
-rw-r--r-- 1 rishitha rishitha 155 Aug 17 16:50 Makefile
-rw-r--r-- 1 rishitha rishitha 0 Aug 17 16:45 README.md
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:51 bin
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:45 include
drwxr-xr-x 2 rishitha rishitha 4096 Aug 17 16:51 src
rishitha@Rishitha:~/system-programming$ pstree -p
systemd(1)─agetty(299)
            └─chronyd-starter(119)─chronyd(239)─chronyd(249)
            └─cron(120)
            └─dbus-daemon(121)
            └─init-systemd(Ub(2))─SessionLeader(834)─Relay(837)(835)─ba+
                │   └─init(9)─{init}(10)
                │   └─login(340)─bash(420)
                │   └─{init-systemd(Ub)}(11)
            └─networkd-dispat(166)
            └─polkitd(555)─{polkitd}(557)
                │   └─{polkitd}(558)
                │   └─{polkitd}(559)
            └─rsyslogd(189)─{rsyslogd}(229)
                │   └─{rsyslogd}(230)
                │   └─{rsyslogd}(231)
            └─systemd(391)─(sd-pam)(395)
            └─systemd-journal(55)
            └─systemd-logind(173)
            └─systemd-resolve(87)
            └─systemd-udev(97)
            └─unattended-upgr(296)─{unattended-upgr}(304)
rishitha@Rishitha:~/system-programming$ ps -ef --forest
UID      PID     PPID    C  STIME TTY          TIME CMD
root      1       0  0 16:39 ?        00:00:01 /sbin/init
root      2       1  0 16:39 hvc0     00:00:00 /init
root      9       2  0 16:39 hvc0     00:00:00 \_ plan9 --control-sock
root     340     2  0 16:39 ?        00:00:00 \_ login -- rishitha
rishitha 420     340  0 16:39 pts/1    00:00:00 | \_ -bash
root     834     2  0 16:46 ?        00:00:00 \_ /init
root     835     834  0 16:46 ?        00:00:00 \_ /init
rishitha 837     835  0 16:46 pts/0    00:00:00 \_ -bash
rishitha 1333    837  0 16:52 pts/0    00:00:00 \_ ps -ef -
root      55       1  0 16:39 ?        00:00:00 /usr/lib/systemd/systemd
systemd+  87       1  0 16:39 ?        00:00:00 /usr/lib/systemd/systemd
root      97       1  0 16:39 ?        00:00:01 /usr/lib/systemd/systemd
root     119     1  0 16:39 ?        00:00:00 /bin/sh /usr/lib/systemd
_chrony   239     119  0 16:39 ?        00:00:00 \_ /usr/sbin/chronyd -n
_chrony   249     239  0 16:39 ?        00:00:00 \_ /usr/sbin/chrony
root     120     1  0 16:39 ?        00:00:00 /usr/sbin/cron -f -p
message+ 121     1  0 16:39 ?        00:00:00 @dbus-daemon --system --
root     166     1  0 16:39 ?        00:00:00 /usr/bin/python3 /usr/bi
root     173     1  0 16:39 ?        00:00:00 /usr/lib/systemd/systemd
syslog    189     1  0 16:39 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root     296     1  0 16:39 ?        00:00:00 /usr/bin/python3 /usr/sh
root     299     1  0 16:39 tty1     00:00:00 /usr/sbin/agetty --nres
rishitha 391     1  0 16:39 ?        00:00:00 /usr/lib/systemd/systemd
rishitha 395     391  0 16:39 ?        00:00:00 \_ (sd-pam)
polkitd   555     1  0 16:42 ?        00:00:00 /usr/lib/polkit-1/polkit
rishitha@Rishitha:~/system-programming$ ps -o pid,ppid,stat,cmd
PID      PPID    STAT  CMD
837      835     Ss    -bash
1343     837     R+    ps -o pid,ppid,stat,cmd
rishitha@Rishitha:~/system-programming$ strace --version

```

[illegible]

[illegible]

